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AquaFetch: A Unified Python Interface for Water Resource Dataset Acquisition and Harmonization

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Software

- Review
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Summary

AquaFetch is a Python package designed for the automated downloading, parsing, cleaning, and harmonization of freely available water resource datasets related to rainfall-runoff processes, surface water quality, and wastewater treatment. The package currently supports 70 datasets, downloading and transforming raw data into consistent, easy-to-use analysis-ready data. This allows users to directly access and utilize the data without labor-intensive and time-consuming preprocessing.

The package comprises three submodules, each representing a different type of water resource data: `rr` for rainfall-runoff processes, `wq` for surface water quality, and `wwt` for wastewater treatment. The `rr` submodule offers data for 47,716 catchments worldwide, encompassing both dynamic and static features for each catchment. The dynamic features consist of observed streamflow and meteorological time series, averaged over the catchment area, available at daily or hourly time steps. Static features include constant parameters such as land use, soil, topography, and other physiographical characteristics, along with catchment boundaries. This submodule not only provides access to established rainfall-runoff datasets such as CAMELS and LamaH but also introduces new datasets compiled for the first time from publicly accessible online data sources. The `wq` submodule offers access to 16 surface water quality datasets, each containing various water quality parameters measured across different spaces and times. The `wwt` submodule provides access to 22,471 experimental measurements related to wastewater treatment techniques such as adsorption, photocatalysis, and sonolysis.

The development of AquaFetch was inspired by the growing availability of diverse water resource datasets in recent years. As a community-driven project, the codebase is structured to allow contributors to easily add new datasets, ensuring the package continues to expand and evolve to meet future needs.

Statement of need

In recent years, there has been a significant increase in the release of datasets across various domains, including water resources. This surge is driven by advancements in computational and storage technologies, as well as the growing need to develop robust, accurate data-driven solutions to address challenges such as climate change, water scarcity, and environmental pollution. As a result, a wealth of national and global spatio-temporal datasets has become freely accessible online. These datasets are invaluable for applications like flood forecasting, climate change analysis, aquatic ecosystem management, improving drinking water safety, and optimizing wastewater treatment processes.

Despite the availability of these datasets, importing them into Python remains cumbersome. Researchers must often sift through multiple sources, including search engines, GitHub repositories, and various websites, to locate the necessary data. The diversity of data providers

means datasets are frequently presented in inconsistent units and stored in varying formats. Additionally, many datasets require extensive preprocessing before they can be used for analysis or modeling. This makes acquiring, cleaning, organizing, and managing data a complex task requiring advanced data handling skills.

These challenges highlight the need for a unified, consistent, automated, and reusable framework for extracting hydrological and environmental data. The AquaFetch package addresses this gap by leveraging data-handling tools such as Pandas (McKinney, 2010), NumPy (Harris et al., 2020), xarray (Hoyer & Hamman, 2017), and Shapely to offer a streamlined workflow for automatic data extraction from multiple sources in various formats.

Several other packages also aim to facilitate the acquisition and processing of water resource data. For example, tools like dataretrieval (Hodson & Horsburgh, 2023), hydrofunctions (Roberge, 2018), and harmonize-wq (Bousquin & Mullin, 2024) provide streamlined access to hydrological and water quality data, though they are primarily limited to data from U.S. Geological Survey (USGS) and Environmental Protection Agency (EPA) stations. Data Retriever (Senyondo et al., 2017) offers access to over 200 parameters, including water quality, but lacks comprehensive rainfall-runoff datasets and wastewater treatment data. Meanwhile, tsp (Brown, 2022) focuses on managing ground temperature data, and the HyRiver suite (Taher Chegini1 & Leung, 2021) provides access to hydro-climate data exclusively from USGS's National Water Information Service (NWIS). Geodata-Harvester (Haan et al., 2023) offers reusable and automated workflows for extracting diverse geospatial and environmental data for Australia.

In the R programming language, packages like weathercan (LaZerte1 & Albers, 2018), weatherOz (Pires et al., 2024), and riversCentralAsia (Marti & Siegfried, 2023) provide hydro-climate data for regions such as Canada, Australia, and Central Asia, respectively. However, these packages often lack the static catchment features essential for modeling rainfall-runoff processes and do not offer comprehensive datasets for this purpose.

The Caravan initiative (Nascimento et al., 2023) stands out as a global platform for hydrometeorological data with a distinct focus. While it aims to create large-sample hydrology datasets globally, AquaFetch facilitates access to a broad range of data via a unified interface, including processed rainfall-runoff datasets from countries like Japan, Thailand, Ireland, and Poland — regions not covered by Caravan. Additionally, Caravan emphasizes cloud-based processing using the proprietary Google Earth Engine and does not provide a standalone Python package, requiring users to upload catchment boundaries for extracting static features. This highlights the distinct operational focuses of Caravan and AquaFetch.

Ultimately, we hope this package will foster the development of benchmark datasets in hydrological and environmental sciences, enhancing the comparability and reproducibility of data-driven solutions for water resource management.

Implementation and Archiecture

The AquaFetch package is structured using both functional and Object-Oriented Programming (OOP) designs. The OOP design is employed for handling more complex datasets within the `rr` and `wq` submodules, while simpler datasets are managed through a functional interface. The package's code is logically organized, with `rr`, `wq`, and `wwt` subdirectories present in both the source code (aqua_fetch) and tests directories. All public classes and functions are accessible from the parent directory, allowing for straightforward imports as shown below:

```
from aqua_fetch import RainfallRUnoff
from aqua_fetch import SWatch
from aqua_fetch import mg_degradation
```

Datasets are downloaded upon the first call to the respective function or class. The data is saved locally and will not be redownloaded unless the user explicitly requests it to be

overwritten. The package also leverages parallel processing to expedite the downloading and parsing of large datasets in the `rr` submodule, significantly speeding up data retrieval when multiple CPU cores are available.

Rainfall Runoff datasets

Table 1 provides a comprehensive list of the 32 rainfall-runoff datasets currently covered by the package. These datasets include series such as CAMELS (Addor et al., 2017) and LamaH (Klingler et al., 2021), published over the last decade and offering data for specific countries or regions. The GRDC-Caravan is the only dataset providing data from catchments globally (Figure 1). These datasets consist of three types of data: time series, tabular, and catchment shapefiles. The static catchment features are presented as tabular data, where each row corresponds to one catchment and each column to one static feature. The dynamic features of catchments include time series of meteorological data and observed streamflow, available at daily or hourly timesteps.

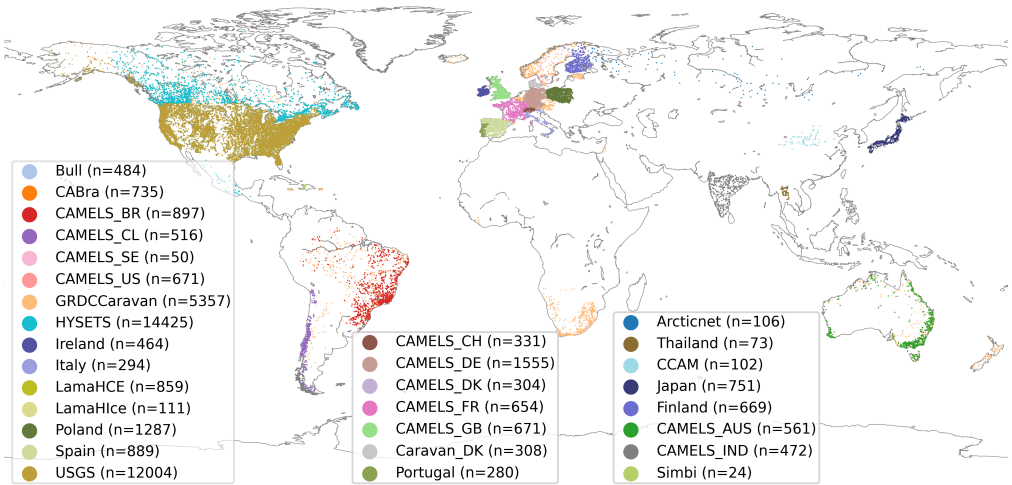

Figure 1: Locations of catchment gauge stations covered by each of the 32 rainfall-runoff datasets

In addition to published datasets, this package introduces 10 new datasets for rainfall-runoff modeling. These datasets have not yet been published but follow the CAMELS dataset series convention. They include Ireland, Finland, Italy, Poland, Portugal, Japan, Thailand, Arcticnet, Spain, and the USGS. The observed streamflow data are sourced from the national meteorological or hydrological websites of the respective countries. Catchment boundaries and meteorological data for Ireland, Finland, Italy, Poland, and Portugal are obtained from EStreams (Nascimento et al., 2024), and similarly for Japan, Thailand, Arcticnet, and Spain from GSHA (Yin et al., 2024). For USGS, the catchment boundaries are sourced from HYSETS (Arsenault et al., 2020).

All datasets listed in Table 1 are accessible via the `RainfallRunoff` class, which allows for a unified and consistent approach to each dataset. The class provides several methods to access static features, dynamic features, or catchment boundaries. Although the raw data files for each dataset may come in different formats, the methods to access these features through the `RainfallRunoff` class remain the same. Individual classes for each dataset are also available and may offer more control to users over specific datasets. However, for most cases, the use of the `RainfallRunoff` class will suffice.

The naming and units of dynamic features in each dataset may vary. However, we have

standardized these features using the formula `name_unit_specifier` for each dynamic feature across all datasets. In this formula, the specifier can indicate the source (such as ERA5 or MSWEP for precipitation), the method used to calculate the feature (like makkink or penman for evapotranspiration), or the aggregation type (min, max, mean). For example, a precipitation dynamic feature from MSWEP would be labeled as pcp_mm_mswep. This approach ensures that feature names are representative and understandable. Dynamic features for which this method is inapplicable retain their original names.

Another feature of the AquaFetch is the optional inclusion of static and dynamic features from EStreams and GSHA for all datasets listed in Table 1. This is beneficial as EStreams and GSHA include several static and dynamic features calculated for the catchments, which are not included in other datasets. For instance, EStreams provides information on annual variation in land use for all European catchments, a feature not available in CAMELS-GB (Coxon et al., 2020) or other European datasets. This step is optional since it initiates the download of GSHA and EStreams datasets which can be time-consuming and may not always be necessary.

Duplicate datasets

Certain datasets in our package feature overlapping stations from the same region. For example, both the Bull (Senent-Aparicio et al., 2024) and Spain datasets both cover Spain. However, the Bull dataset was introduced by by Aparicio et al. (2024) (Senent-Aparicio et al., 2024), whereas the Spain dataset was introduced in our work. The Spain dataset contains more stations, totaling 889, while the Bull dataset includes 484 stations. Similarly, both the CABra (Almagro et al., 2021) and CAMELS_BR (Chagas et al., 2020) datasets cover Brazil and have been published in peer-reviewed journals. However, they differ in their temporal coverage and the number of static and dynamic features. Furthermore, Denmark is covered by two datasets, Caravan_DK (Koch, 2023) and CAMELS_DK (Liu et al., 2024), which differ in temporal coverage and the number of static and dynamic features. The HYSETS dataset (Arsenault et al., 2020) covers Mexico, the US, and Canada. However, we identified issues with the observed streamflow data for the US in HYSETS. As a result, we introduced the USGS class, which focuses specifically on the US region. The catchment boundaries, static features, and meteorological data for USGS, however, are still obtained from HYSETS.

Table 1: Summary of rainfall-runoff datasets covered in AquaFetch package

Name	Daily stations	Hourly stations	Dynamic features	Static features	Temporal Coverage	Spatial Coverage	Reference
Arcticnet	106		27	35	1979–2003	Arctic (Russia)	R-Arcticnet
Bull	484		55	214	1990–2020	Spain	Aparicio et al., 2024
CABra	735		12	97	1980–2010	Brazil	Almagro et al., 2021
CAMELS_AUS	222, 561		26	166, 187	1900–2018	Australia	Flower et al., 2021
CAMELS_BR	897		10	67	1920–2019	Brazil	Chagas et al., 2020
CAMELS_CH	331		9	209	1981–2020	Switzerland, Austria, France, Germany Italy	Hoege et al., 2023
CAMELS_CL	516		12	104	1913–2018	Chile	Alvarez-Garreton et al., 2018

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Name	Daily stations	Hourly stations	Dynamic features	Static features	Temporal Coverage	Spatial Coverage	Reference
CAMELS_DE	1555		21	111	1951–2020	Germany	Loritz et al., 2024
CAMELS_DK	304		13	119	1989–2023	Denmark	Liu et al., 2024
CAMELS_FR	654		22	344	1970–2021	France	Delaigue et al., 2024
CAMELS_GB	671		10	145	1970–2015	Britain	Coxon et al., 2020
CAMELS_IND	472		20	210	1980–2020	India	Mangukiya et al., 2024
CAMELS_SE	50		4	76	1961–2020	Sweden	Teutschbein et al., 2024
CAMELS_US	671		8	59	1980–2014	USA	Newman et al., 2014
Caravan_DK	308		38	211	1981–2020	Denmark	Koch, J. (2022)
CCAM	102		16	124	1990–2020	China	Hao et al., 2021
Finland	669		27	35	2012–2023	Finland	Nascimento et al., 2024
GRDCCaravan	5357		39	211	1950–2023	Global	Faerber et al., 2023
HYSETS	14425		5	28	1950–2018	North America	Arsenault et al., 2020
Italy	294		27	35	1992–2020	Italy	Nascimento et al., 2024
Ireland	464		27	35	1992–2020	Ireland	EPA Ireland
Japan	751	696	27	35	1979–2022	Japan	river.go.jp
LamaHCE	859	859	22	80	1981–2019	Central Europe	Klingler et al., 2021
LamaHIce	111	111	36	154	1950–2021	Iceland	Helgason and Nijssen 2024
Poland	1287		27	35	1992–2020	Poland	Nascimento et al., 2024
Portugal	280		27	35	1992–2020	Portugal	SNIRH Portugal
RRLuleaSweden	1		2	0	2016–2019	Lulea (Sweden)	Broekhuizen et al., 2020
Spain	889		27	35	1979–2020	Spain	ceh-flumen64
Simbi	70		3	232	1920–1940	Haiti	Bathelemy et al., 2024
Thailand	73		27	35	1980–1999	Thailand	RID project
USGS	12004	1541	5	27	1950–2018	USA	USGS nwis
WaterBenchIowa	125		3	7	2011–2018	Iowa (USA)	Demir et al., 2022

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Water Quality datasets

The wq submodule contains datasets that represent surface water chemistry at various locations worldwide. Currently, it includes 16 water quality datasets, but we anticipate this number will increase in the future. The spatial and temporal coverage of these datasets are detailed in Table 2 while the exact location of measuring stations are depicted in Figure 2.

Table 2: Summary of water quality datasets covered in AquaFetch package

Name	Variables Covered	Temporal Coverage	Spatial Coverage	Reference
SWatCh	24	1960–2022	Global	Lobke et al., 2022
GRQA	42	1898–2020	Global	Virro et al., 2021
Quadica	10	1950–2018	Germany	Ebeling et al., 2022
RC4USCoast	21	1850–2020	USA	Gomez et al., 2022
Busan Beach	14	2018–2019	Busan (South Korea)	Jang et al
Ecoli Mekong River	10	2011–2021	Mekong river (Houay Pano)	Boithias et al., 2022
Ecoli Mekong River (Laos)	10	2011–2021	Mekong River (Laos)	Boithias et al., 2022
Ecoli Houay Pano (Laos)	10	2011–2021	Houay Pano (Laos)	Boithias et al., 2022
CamelsChem	18	1980–2018	Conterminous USA	Sterle et al., 2024
Sylt Roads	14	1973–2019	Continental USA	Rick., 2023
SanFranciscoBay	8	1969–2015	San Francisco (USA)	Sterle et al., 2024
White Clay Creek	2	2001–2012	White Creek (USA)	Hydroshare
Selune River	5	2021–2022	Selune River (France)	Moustapha Ba et al., 2023
Buzzards Bay	64	1992–2018	Buzzards Bay (USA)	Jakuba et al., 2021
GRiMeDB	18	-	Global	Stanley et al., 2023
River Chemistry Siberia	30	1991–2012	Siberia	Liu et al., 2022

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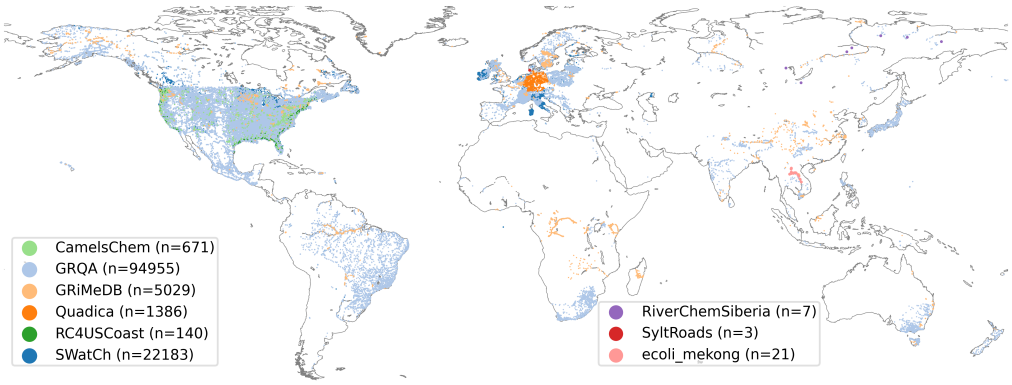

Figure 2: Locations of measuring stations of water quality datasets

Wastewater treatment datasets

The wwt submodule contains data from approximately 22,471 experiments focused on the removal of various contaminants from wastewater using treatment strategies such as adsorption, photocatalysis, and sonolysis. This submodule provides a unified interface to access all this data, which is scattered across the literature, in a standardized format using a few Python functions. It is important to note that we do not introduce this data since this data has already been utilized and analyzed in various peer-reviewed scientific publications. However, we offer a simple and easy-to-use interface to access this existing data. The availability of such a large corpus of experimental data can significantly aid in data-driven modeling and material discovery. A summary of these datasets is provided in Table 3.

Table 3: Summary of wastewater treatment datasets covered in the package

Treatment Process	Parameters	Target Pollutant	Data Points	Reference
Adsorption	26	Emerg. Contaminants	3,757	Jaffari et al., 2023
Adsorption	15	Cr	219	Ishtiaq et al., 2024
Adsorption	30	Heavy Metals	1,518	Jaffari et al., 2023
Adsorption	30	PO4	5,014	Iftikhar et al., 2024
Adsorption	12	Industrial Dye	1,514	Iftikhar et al., 2023
Adsorption	17	Heavy Metals	689	Shen et al., 2023
Adsorption	8	P	504	Leng et al., 2024
Adsorption	8	N	211	Leng et al., 2024
Adsorption	13	As	1,605	Huang et al., 2024
Photocatalysis	11	Melachite Green	1,200	Jaffari et a., 2023
Photocatalysis	23	Dyes	1,527	Kim et al., 2024
Photocatalysis	15	2,4,Dichlorophenoxy-acetic acid	1,044	Kim et al., 2024
Photocatalysis	25	Multiple	2,078	GitHub
Photocatalysis	8	Tetracycline	374	Abdi et al., 2022

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Treatment Process	Parameters	Target Pollutant	Data Points	Reference
Photocatalysis	7	TiO2	446	Jiang et al., 2020
Photocatalysis	8	Multiple	457	Jiang et al., 2020
Sonolysis	6	Cyanobacteria	314	Jaffari et al., 2024

Testing and dependencies

The AquaFetch code is hosted on GitHub at https://github.com/hyex-research/AquaFetch. The package's core dependencies include requests (Rogozhnikov, 2018), NumPy, and Pandas. xarray is utilized for saving data in netCDF5 (NSF Unidata et al., n.d.) format, which is efficient for handling large datasets; however, this step is an optional dependency. Other optional dependencies include Matplotlib (Hunter, 2007) for plotting and visualization, openpyxl (Gazoni & Clark, 2024) for parsing Microsoft Excel files, along with Shapely, Shapefile, and Fiona for processing shapefiles. The documentation for the package is available on ReadTheDocs at https://aquafetch.readthedocs.io, featuring several tutorials that outline its usage and capabilities.

Adhering to the 'unit test' protocol, comprehensive testing has been implemented for all data classes and functions. Since downloading the datasets is time-consuming, the tests are conducted offline under the assumption that the datasets are already downloaded. These unit tests verify the number and types of parameters returned by the data functions.

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